

The dimension of moment of force (torque) is,

$$\begin{aligned}\tau &= \text{distance} \times \text{force} \\ &= [L^1] [M^1 L^1 T^{-2}] \\ &= [M^1 L^2 T^{-2}]\end{aligned}$$

The dimension of surface tension is,

$$\begin{aligned}S &= \frac{\text{force}}{\text{length}} \\ &= \frac{[M^1 L^1 T^{-2}]}{[L^1]} \\ &= [M^1 T^{-2}]\end{aligned}$$

The dimension of modulus of elasticity is,

$$\begin{aligned}E &= \frac{\text{Stress}}{\text{Strain}} = \frac{\left(\frac{\text{force}}{\text{area}}\right)}{\text{strain}} \\ &= \frac{\left(\frac{[M^1 L^1 T^{-2}]}{[L^2]}\right)}{[M^0 L^0 T^0]} \\ &= [M^1 L^{-1} T^{-2}]\end{aligned}$$

The dimension of the coefficient of viscosity is,

$$\eta = \frac{\text{force} \times \text{length}}{\text{area} \times \text{velocity}}$$

1

✓

Next →

The dimension of moment of force (torque) is,

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The dimension of the coefficient of viscosity is,

$$\eta = \frac{\text{force} \times \text{length}}{\text{area} \times \text{velocity}}$$

1

✓

Next →

DIDN'T ATTEMPT 

21.23%

GOT IT RIGHT 

88.72%

GOT IT WRONG 

11.28%

SOLUTION

 Video Solution

$$T_{\text{mean}} = \frac{53+52+55+54+51}{5} = 53\text{sec.}$$

$$\left. \begin{array}{l} \Delta T_1 = T_{\text{mean}} - T_1 = 0 \\ \Delta T_2 = 1\text{sec} \\ \Delta T_3 = 2\text{sec} \\ \Delta T_4 = 1\text{sec} \\ \Delta T_5 = 2\text{sec} \end{array} \right\}$$

$$\begin{aligned} \Delta T_{\text{mean}} &= \frac{0+1+2+1+2}{5} \\ &= \frac{6}{5} = 1.2 \text{ sec.} \end{aligned}$$

 Prev

2

v

Next 

By third's law of motion

$$v^2 = u^2 + 2as$$

$$v^2 - u^2 = -2 \left(\frac{F}{m} \right) \cdot s$$

$$-u^2 = -2 \frac{F}{m} \cdot s$$

$$u^2 = \frac{2F \cdot s}{m}$$

$$\text{or } m = \frac{2F \cdot s}{u^2} \Rightarrow m \propto s$$

Given $s_1 = x$

$$s_2 = ?$$

$$m_1 = m$$

$$m_2 = m \times 25\% + m$$

$$\text{Now, } \frac{m_1}{m_2} = \frac{s_1}{s_2}$$

$$\frac{m}{m \times \frac{25}{100} + m} = \frac{x}{s_2}$$

$$s_2 = \frac{x \times m \left(\frac{100+25}{100} \right)}{m}$$

$$s_2 = \frac{125}{100} x$$

$$s_2 = 1.25x$$

← Prev

3

✓

Next →

Incorrect

You: 153 sec
Others: 119 sec

120 sec

DIFFICULTY
MediumCHAPTER
Laws of MotionTOPIC
Non-uniform Circular MotionDIDN'T ATTEMPT ☐
22.27%GOT IT RIGHT ☒
64.41%GOT IT WRONG ☐
35.59%

SOLUTION

[▶ Video Solution](#)Tension between P and Q is

$$T_1 = \text{centripetal force on } Q = mr\omega^2$$
$$= 200 \times 1 \times \omega^2 \left(\text{g} \cdot \text{m} \cdot \frac{\text{rad}^2}{\text{s}^2} \right)$$

Tension between O and P is

$$T_2 = \text{centripetal force on } Q + \text{centripetal force on } P$$
$$= 200 \times 1 \times \omega^2 + 200 \times \frac{1}{2} \times \omega^2$$
$$= 300 \times 1 \times \omega^2 \left(\text{g} \cdot \text{m} \cdot \frac{\text{rad}^2}{\text{s}^2} \right)$$

Ratio of tension is

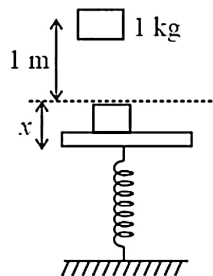
$$\frac{T_1}{T_2} = \frac{200 \times 1 \times \omega^2}{300 \times 1 \times \omega^2} = \frac{2}{3}$$

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v

[Next >](#)Activate Windows
Go to Settings to activate Windows.



In case of maximum compression, total mechanical energy will be conserved.

$$TE_1 = TE_2$$

Assuming dotted line to be reference for gravitational potential energy.

$$\Rightarrow mgh + 0 = mg(-x) + \frac{1}{2} kx^2$$

$$\Rightarrow 1 \times 10 \times 1 = 1 \times 10 \times (-x) + \frac{1}{2} \times 15 \times x^2$$

$$\Rightarrow 3x^2 - 4x - 4 = 0$$

$$\Rightarrow x = 2 \text{ m}$$

From the body of mass m , the distance of the point where gravitational field is zero

$$x = \frac{d}{\sqrt{\frac{m_2}{m_1}} + 1}$$
$$\Rightarrow x = \frac{r}{\sqrt{\frac{9m}{m}} + 1}$$

$$\therefore x = \frac{1}{4}r$$

$$\text{Again } (r - x) = r - \frac{r}{4} = \frac{3r}{4}$$

$\therefore$ Potential, $v = v_1 + v_2$

$$= \frac{Gm_2}{X} - \frac{Gm_1}{r - x} = -\frac{Gm}{r/4} - \frac{G(9m)}{3r/4}$$
$$= -\frac{4Gm}{r} - \frac{12Gm}{r} = -\frac{16Gm}{r}$$

Please select the reason why you attempted the question wrong, this will help us in providing more accurate analysis.



Misinterpreted Question



Forgot Formula



Overthinking



Missed a Step

EVALUATION

Incorrect

TIME TAKEN

You: 131 sec
Others: 69 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Hard

CHAPTER

Thermodynamics

TOPIC

First Law of Thermodynamics

DIDN'T ATTEMPT ☐

3.74%

GOT IT RIGHT ☒

56.32%

GOT IT WRONG ☒

43.68%

SOLUTION

[▶ Video Solution](#)

(i) At constant temperature $P \propto \rho$

(ii) At constant pressure $T \propto \frac{1}{\rho}$

(iii) $V \rightarrow \text{constant}$

$\therefore \rho \rightarrow \text{constant}$

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▼

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EVALUATION



TIME TAKEN

You: 222 sec

Others: 99 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Kinetic Theory of Gases

TOPIC

Degree of Freedom and Specific Heat

DIDN'T ATTEMPT ☐

27.59%

GOT IT RIGHT ☒

83.31%

GOT IT WRONG ☐

16.69%

SOLUTION

[▶ Video Solution](#)

Relation of mean free path with temperature and pressure:

$$\lambda \propto \frac{T}{P}$$

At constant temperature the relation of mean free path and pressure is:

$$\frac{\lambda_2}{\lambda_1} = \frac{P_1}{P_2}$$
$$\lambda_2 = 60 \times \frac{1}{100} = 6 \times 10^{-1} \text{ cm}$$

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8

v

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Given, mass of hammer, $m = 200 \text{ kg}$,

steel block of the mass $= 200 \text{ g} = 0.2 \text{ kg}$ and specific

heat capacity of steel, $s = 460 \text{ J kg}^{-1} \text{ K}^{-1}$

Velocity of hammer, $v = 8 \text{ ms}^{-1}$

As we know that,

$$\text{Kinetic energy, KE} = \frac{1}{2} mv^2$$

Putting the given values, we get

$$= \frac{1}{2} \times 200 \times 8^2 = 6400 \text{ J}$$

Hence, the 23% of this energy is converted to heat.

$$\Rightarrow H = \frac{6400 \times 23}{100} = 1472 \text{ J}$$

The rise in temperature of steel,

$$\Delta T = \frac{H}{ms} = \frac{1472}{460 \times 0.2} = 16 \text{ K}$$

Hence, the rise in temperature is 16 K.

← Prev

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✓

Next →

Given, mass of hammer, $m = 200 \text{ kg}$,

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← Prev

9

✓

Next →

EVALUATION

Not Answered

TIME TAKEN

You: 8 sec

Others: 160 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Oscillations

TOPIC

Simple Harmonic Motion

DIDN'T ATTEMPT ☐

14.14%

GOT IT RIGHT ☒

96.25%

GOT IT WRONG ☐

3.75%

SOLUTION

[▶ Video Solution](#)

As per question,

$$|A\omega^2| - |A\omega| = 4 \quad [\because A = 2 \text{ m}]$$

$$2\omega^2 - 2\omega - 4 = 0$$

$$\Rightarrow \omega = 2 \text{ rad/s} \Rightarrow \frac{2\pi}{T} = 2$$

$$\Rightarrow T = \pi = \frac{22}{7} \text{ s}$$

$$\text{Velocity, } v = \omega \sqrt{A^2 - y^2} = 2 \sqrt{(2)^2 - (1)^2} = 2\sqrt{4-1} = 2\sqrt{3} \text{ ms}^{-1}$$

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v

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DIDN'T ATTEMPT ☐
11.67%

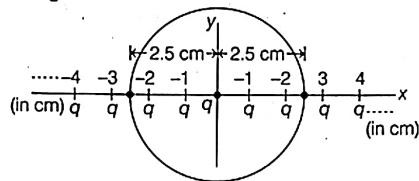
GOT IT RIGHT ☒
92.09%

GOT IT WRONG ☐
7.91%

SOLUTION

[▶ Video Solution](#)

The given situation is shown below



From the above figure total charge enclosed by the sphere of radius $2.5 \text{ cm} = 5q$

$\therefore$ According to gauss law, total electric flux passing through spherical surface,

$$\phi = \frac{\text{total charge enclosed}}{\epsilon_0}$$

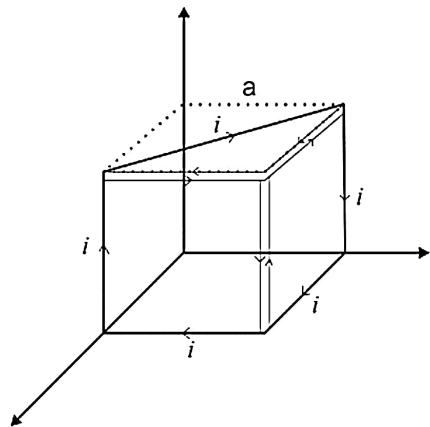
$$= \frac{5q}{\epsilon_0}$$

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▼

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Here $\vec{M} = -ia^2\hat{i} - i\frac{a^2}{2}\hat{j} - ia^2\hat{k}$

$$\vec{M} = -ia^2\left(\hat{i} + \frac{\hat{j}}{2} + \hat{k}\right) \Rightarrow |\vec{M}| = \frac{3}{2}ia^2$$

$$= \frac{3}{2} \times 3 \times 1 \quad |\vec{M}| = \frac{9}{2} \text{ ampere-meter}^2$$

EVALUATION



DIFFICULTY

Medium

DIDN'T ATTEMPT ☐

3.69%

TIME TAKEN

You: 135 sec

Others: 57 sec

CHAPTER

Alternating Current

GOT IT RIGHT ☒

91.78%

TIME ALLOTTED

120 sec

TOPIC

AC Circuits

GOT IT WRONG ☐

8.22%

SOLUTION

[▶ Video Solution](#)

The current in DC circuit containing resistance and capacitance is given by,

$$i = i_0 \left(1 - e^{-\frac{t}{RC}} \right)$$

As the power of e should be dimensionless, the dimensions of RC should be equal to that of time.

;

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EVALUATION



DIFFICULTY

Easy

DIDN'T ATTEMPT ☐

33.55%

TIME TAKEN

You: 235 sec

Others: 174 sec

CHAPTER

Electromagnetic Waves

GOT IT RIGHT ☒

90.79%

TIME ALLOTTED

120 sec

TOPIC

Electromagnetic Waves and Maxwell's equations

GOT IT WRONG ☐

9.21%

SOLUTION

[▶ Video Solution](#)

$$\begin{aligned}
 V_1 &= \frac{\omega}{k} & V_2 &= \frac{\omega}{3k} \\
 \Rightarrow 3 V_2 &= V_1 \\
 3 \frac{1}{\sqrt{\mu_{r2} \epsilon_{r2}}} &= \frac{1}{\sqrt{\mu_{r1} \epsilon_{r1}}} \\
 \Rightarrow \frac{\epsilon_{r2}}{\epsilon_{r1}} &= 9 \frac{\mu_{r1}}{\mu_{r2}} = \frac{9}{4} = 2.25
 \end{aligned}$$

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DIFFICULTY

Hard

CHAPTER

Ray Optics

TOPIC

Lens Formula, Refraction of light from curved surface

DIDN'T ATTEMPT ☐

54.73%

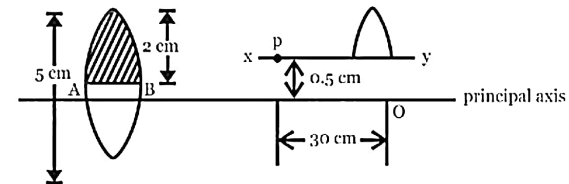
GOT IT RIGHT ☒

66.90%

GOT IT WRONG ☐

33.10%

SOLUTION

[▶ Video Solution](#)

$$\frac{h_1}{h_o} = \frac{f}{u+f}$$

$$h_o = \frac{20}{(-30)+20} \times 0.5$$

$$h_o = -1 \text{ cm}$$

so image will be formed 1.5 cm below xy line.

^

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v

[Next →](#)



4.69% marked this



51.92% marked this

EVALUATION



TIME TAKEN

You: 57 sec

Others: 33 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

Waves and Sound

TOPIC

Interference and Superposition of Waves

DIDN'T ATTEMPT

26.11%

GOT IT RIGHT

51.92%

GOT IT WRONG

48.08%

SOLUTION

Video Solution

Quality (Wave form) of sound distinguish the different sources of sound from each other

So, assertion is false but, reason is true.

:

Prev

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Next

As the total number of wavelengths emitted is 6, let the final excited state of the be n , then

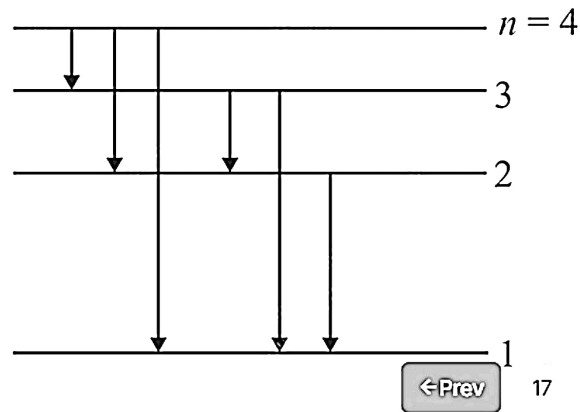
$$\frac{n(n-1)}{2} = 6 \Rightarrow n = 4$$

Hence, the final excited state for the atom is $n = 4$

If the initial state were $n = 3$, in the emission spectrum, then no wavelengths shorter than λ_0 would have occurred. But if we take the initial state to be $n = 2$, then the transition from $n = 2$ to $n = 3$ and from $n = 3$ to $n = 4$ have wavelengths shorter than λ_0 .

Lyman series is the spectral series that occurs when transition takes place from $n \geq 2$ to $n = 1$, therefore in this case, we have 3 spectral lines in Lyman series.

(A), (B) and (D)



Next →

DIDN'T ATTEMPT ☐
7.09%

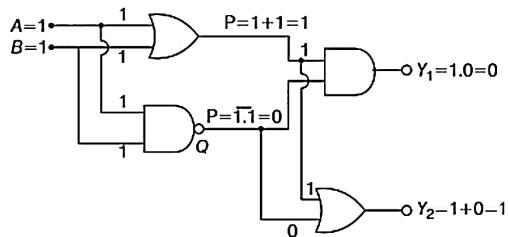
GOT IT RIGHT ☒
89.19%

GOT IT WRONG ☐
10.81%

SOLUTION

[▶ Video Solution](#)

The logical circuit can be explained as,



So, $Y_1 = 0$ and $Y_2 = 1$

Hence, the correct option is (c).

:

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✓

[Next →](#)



Correct

You Marked

79.10% marked this



Neither dark nor bright

7.56% marked this

EVALUATION

Correct

TIME TAKEN

You: 107 sec

Others: 56 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Medium

CHAPTER

Wave Optics

TOPIC

Youngs Double Slit Experiment

DIDN'T ATTEMPT ☐

19.80%

GOT IT RIGHT ☒

79.10%

GOT IT WRONG ☐

20.90%

SOLUTION

[▶ Video Solution](#)

Path difference, $x = (SS_1 + S_1O) - (SS_2 + S_2O)$.

If, $x = n\lambda$, then central fringe at O will be bright.

If, $x = (2n - 1)\lambda/2$, the central fringe at O will be dark.

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You: 198 sec
Others: 115 sec

90 sec

DIFFICULTY

Easy

CHAPTER

Electrostatics

TOPIC

Electric Potential and Potential Energy

DIDN'T ATTEMPT ☐

7.00%

GOT IT RIGHT ☒

92.28%

GOT IT WRONG ☐

7.72%

SOLUTION

[▶ Video Solution](#)

By the principle of superposition, we have

$$V = \sum \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

$$V(A) = \frac{2}{4\pi\epsilon_0} \cdot \frac{q}{L} - \frac{2}{4\pi\epsilon_0} \cdot \frac{q}{\sqrt{5}L}$$

$$V(A) = \frac{2q}{4\pi\epsilon_0 L} \left[1 - \frac{1}{\sqrt{5}} \right]$$

!

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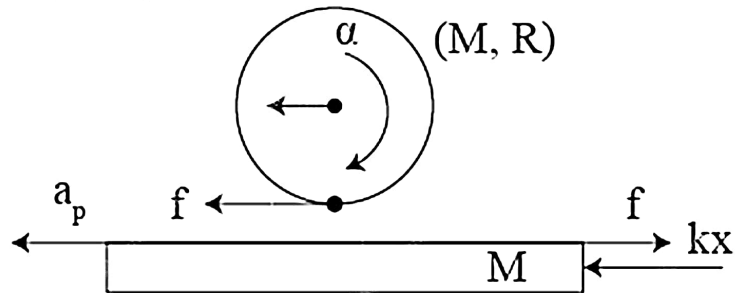
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SOLUTION

[▶ Video Solution](#)

Using no Slipping condition:



$$a_c + \alpha R = a_p$$

$$\frac{f}{M} + \frac{2fR}{MR^2} R = \frac{kx-f}{M}$$

$$\Rightarrow 4f = kx$$

$$x_{\max} = \frac{4\mu Mg}{k}$$

$$\Rightarrow x_{\max} = \frac{4 \times 0.4 \times 2 \times 10}{200} = 0.16 \text{ m} = 16 \text{ cm}$$

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v

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SOLUTION

[▶ Video Solution](#)

$$Y = \frac{\frac{F}{A}}{\frac{l}{L}} = \frac{F \times L}{A \times l} \quad [l = \text{Change in length, } L = \text{Original length}] \text{ (where } Y \text{ is Young's modulus of elasticity)}$$

Since Y , L and A remain same.

$$\Rightarrow F \propto l$$

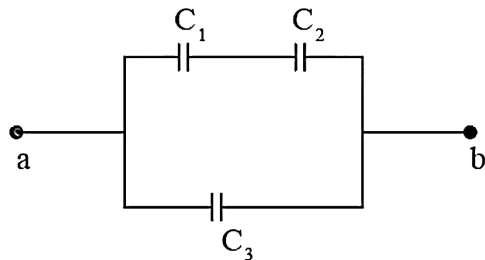
$$\Rightarrow \frac{F_1}{F_2} = \frac{l_1}{l_2}$$

$$\Rightarrow \frac{F}{F_2} = \frac{2 \times 10^{-3}}{4 \times 10^{-3}}$$

$$\Rightarrow F_2 = 2F$$

$$\Rightarrow F_2 = nF$$

$$\Rightarrow n = 2$$



$$C_1 \equiv C_2 = C_3 = C$$

Capacitor C_1 & C_2 are in series.

$$C_{12} = \frac{C_1 C_2}{C_1 + C_2} = \frac{C^2}{2C} = \frac{C}{2}$$

Now C_{12} & C_3 are in parallel,

$$\begin{aligned} \therefore C_T &= C_{12} + C_3 \\ &= \frac{C}{2} + C = \frac{3C}{2} \end{aligned}$$

$$C \text{ is given as } C = \frac{A\epsilon_0}{d}$$

$\therefore$ Capacitance of system,

$$C = \frac{3 A \epsilon_0}{2 d}$$

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23

∨

Next →

Your Answer

1 **Correct**

Correct Answer

1

EVALUATION

Correct

TIME TAKEN

You: 153 sec

Others: 105 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Current Electricity

TOPIC

Electric Current and Drift of Electrons

DIDN'T ATTEMPT ☐

16.45%

GOT IT RIGHT ☒

95.49%

GOT IT WRONG ☒

4.51%

SOLUTION

[▶ Video Solution](#)

Since current $I = neAv_d$ through both rods is same $2(n) eAv v_L = n e (2A)v_R$

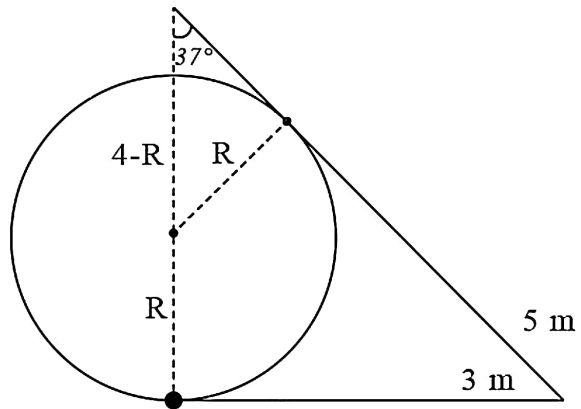
$$\Rightarrow \text{or } \frac{V_L}{V_R} = 1$$

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▼

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$$\sin 37^\circ = \frac{R}{4-R}$$

$$\Rightarrow \frac{3}{5} = \frac{R}{4-R}$$

$$\Rightarrow 12 - 3R = 5R$$

$$\Rightarrow 8R = 12$$

$$\Rightarrow R = \frac{12}{8} = \frac{3}{2}\text{ m}$$

$$V = \frac{qBR}{m} = \frac{1 \times 2 \times 1.5}{1} = 3\text{ m s}^{-1}$$



Decreasing the volume of the vessel to one-half

3.30% marked this



Keeping the temperature of water constant

1.85% marked this

EVALUATION

Not Answered

TIME TAKEN

You: 10 sec

Others: 74 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

Solutions

TOPIC

Vapour pressure and Raoult's law

DIDN'T ATTEMPT ☐

10.30%

GOT IT RIGHT ☒

86.96%

GOT IT WRONG ☐

13.04%

SOLUTION

[▶ Video Solution](#)

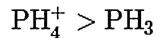
When salt (non-volatile solute) is added to water kept in a sealed vessel, boiling point of the solution will be elevated or increased. So, vapour pressure of water (solution) will be decreased. Lowering of vapour pressure ($\Delta p = p^\circ - p$) is also proportional to mole fraction of the solute (χ_B) present in the solution.

$$\Delta p \propto \chi_B [p^\circ = \text{vapour pressure of pure water } p = \text{vapour pressure of solution}]$$

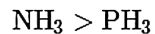
[< Prev](#)

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21.59% marked this



32.81% marked this

EVALUATION

Marked for Review

TIME TAKEN

You: 3 sec

Others: 31 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Medium

CHAPTER

Chemical Bonding and Molecular Structure

TOPIC

Hybridisation and VSEPR Theory

DIDN'T ATTEMPT ☐

25.62%

GOT IT RIGHT ☒

27.95%

GOT IT WRONG ☒

72.05%

SOLUTION

▶ Video Solution

According to drago's rule, in H_2S , bond are formed by p-orbitals 100% p-character while in H_2O bonds are formed by sp^3 (75 % p-character) hybrid orbitals.

◀ Prev

27



Next ▶

Correct

83.02% marked this

10.69% marked this

EVALUATION

Not Answered

TIME TAKEN

You: 7 sec

Others: 95 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Medium

CHAPTER

Aldehydes and Ketones

TOPIC

Chemical reactions for aldehydes and ketones

DIDN'T ATTEMPT ☐

54.04%

GOT IT RIGHT ☒

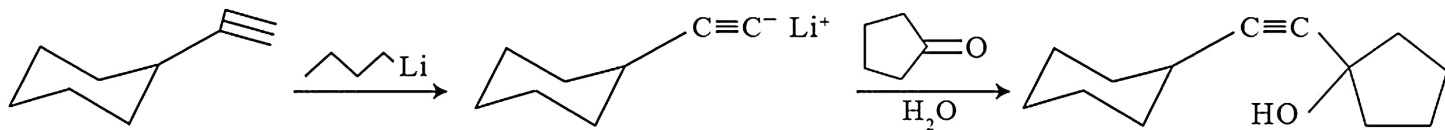
83.02%

GOT IT WRONG ☐

16.98%

SOLUTION

[▶ Video Solution](#)



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✓

Next →

Activate Windows
Go to Settings to activate Windows.

EVALUATION

Marked for Review

DIFFICULTY

Medium

DIDN'T ATTEMPT ☐

11.77%

TIME TAKEN

You: 17 sec

Others: 40 sec

CHAPTER

Chemical Bonding and Molecular Structure

GOT IT RIGHT ☒

87.89%

TIME ALLOTTED

90 sec

TOPIC

Van Der Waals Forces, Hydrogen Bonding and Metallic Bonding

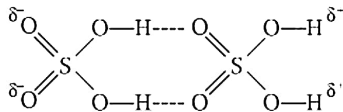
GOT IT WRONG ☒

12.11%

SOLUTION

[▶ Video Solution](#)

H_2SO_4 involves H-bonding due to which it is less volatile.

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29

v

[Next >](#)



0.36% marked this



1.69% marked this

EVALUATION

VIEWED BY REVIEW

TIME TAKEN

You: 9 sec

Others: 44 sec

TIME ALLOTTED

60 sec

DIFFICULTY

Easy

CHAPTER

Classification of Elements and Periodicity in Properties

TOPIC

Periodic properties

DIDN'T ATTEMPT ☐

2.32%

GOT IT RIGHT ☒

93.76%

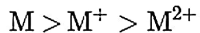
GOT IT WRONG ☐

6.24%

SOLUTION

[▶ Video Solution](#)

In isoelectronic species atomic radius $\propto \frac{1}{Z_{\text{eff}}} \propto \frac{1}{\text{charge on cation}}$



Hence, increasing order of radius is $\text{Ca}^{2+} < \text{K}^+ < \text{Ar}$.

[< Prev](#)

30



[Next >](#)

EVALUATION

Not Answered

TIME TAKEN

You: 10 sec

Others: 32 sec

TIME ALLOTTED

120 sec

DIFFICULTY

Hard

CHAPTER

p Block Elements (Group 13 & 14)

TOPIC

Boron and Its Compounds

DIDN'T ATTEMPT ☐

45.81%

GOT IT RIGHT ☒

35.58%

GOT IT WRONG ☐

64.42%

SOLUTION

[▶ Video Solution](#)

B(OH)_3 → Weak acid
(pK_a can't be negative)

B(OH)_3 → Monobasic acid

B(OH)_3 → H-bonds within a layer
not in between layers.

[← Prev](#)

31

v

[Next →](#)



9.47% marked this



a, c, d

Correct

78.29% marked this



5.82% marked this



a, b, d

6.42% marked this

EVALUATION

Marked for Review

DIFFICULTY

Medium

DIDN'T ATTEMPT ☐

7.39%

TIME TAKEN

You: 2 sec

Others: 105 sec

CHAPTER

Structure of Atom

GOT IT RIGHT ☒

78.29%

TIME ALLOTTED

120 sec

TOPIC

Quantum mechanical model

GOT IT WRONG ☒

21.71%

SOLUTION

▶ Video Solution

a, c, d are correct. Check video solution for detailed explanation.

← Prev

32



Next →

EVALUATION

Marked for Review

TIME TAKEN

You: 4 sec

Others: 77 sec

TIME ALLOTTED

60 sec

DIFFICULTY

Easy

CHAPTER

Electrochemistry

TOPIC

Cells and Electrode Potential, Nernst Equation

DIDN'T ATTEMPT ☐

6.60%

GOT IT RIGHT ☒

90.57%

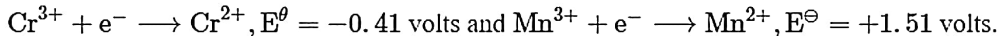
GOT IT WRONG ☐

9.43%

SOLUTION

[▶ Video Solution](#)

The negative value of standard reduction potential (SRP) indicates that the reactant is more stable than product while for positive (SRP) it is vice-versa.



This shows that Cr^{2+} is unstable and has a tendency to acquire more stable Cr^{3+} state by acting as a reducing agent. On the other hand Mn^{3+} is unstable and is reduced to more stable Mn^{2+} state which has d^5 configuration.

[< Prev](#)

33

v

[Next >](#)

EVALUATION

Not Answered

TIME TAKEN

You: 4 sec

Others: 65 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Hard

CHAPTER

Hydrocarbons

TOPIC

Properties and Preparation of Alkynes

DIDN'T ATTEMPT ☐

19.31%

GOT IT RIGHT ☒

57.21%

GOT IT WRONG ☐

42.79%

SOLUTION

[▶ Video Solution](#)

$\text{CH}_3\text{CH}=\text{CH}_2$ to $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$ in the presence

of HCl in a peroxide is not feasible as peroxide/kharash effect is not observed for HCl and HI .

$\text{CH}_3\underset{\text{Cl}}{\text{CH}}-\underset{\text{Cl}}{\text{CH}_2}$ to $\text{CH}_3\text{C}\equiv\text{CH}$ using KOH/Δ is not

feasible as this reaction would give an alkene which when treated with sodamide gives alkyne.

Thus, I and III are not feasible.

[< Prev](#)

34

v

[Next >](#)

98.21% marked this

0.87% marked this

EVALUATION

Marked for review

TIME TAKEN

You: 1 sec
Others: 52 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

d and f Block Elements

TOPIC

Properties of D-block elements

DIDN'T ATTEMPT ☐

2.07%

GOT IT RIGHT ☒

98.21%

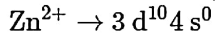
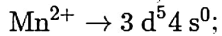
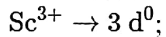
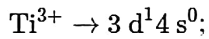
GOT IT WRONG ☐

1.79%

SOLUTION

▶ Video Solution

In Mn^{2+} number of unpaired $d e^- = 5$. So it has maximum magnetic moment according to the formula. $\mu = \sqrt{n(n+2)}$



◀ Prev

35

▼

Next ▶

Activate Windows
Go to Settings to activate Windows.

EVALUATION

Not Answered

TIME TAKEN

You: 3 sec

Others: 32 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Hard

CHAPTER

Coordination Compounds

TOPIC

Werner's theory and Valence Bond Theory

DIDN'T ATTEMPT ☐

44.63%

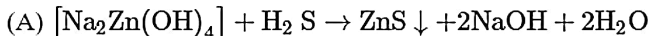
GOT IT RIGHT ☒

20.14%

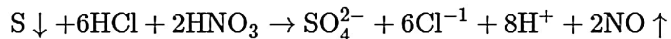
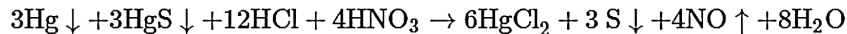
GOT IT WRONG ☒

79.86%

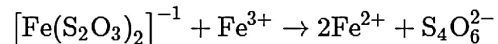
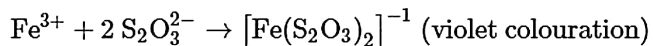
SOLUTION

[▶ Video Solution](#)

(B)



(D)

[< Prev](#)

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v

[Next >](#)

EVALUATION

Marked for Review

TIME TAKEN

You: 1 sec
Others: 39 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Easy

CHAPTER

p Block Elements (Group 15, 16, 17 & 18)

TOPIC

Oxygen and Its Compounds

DIDN'T ATTEMPT ☐

11.13%

GOT IT RIGHT ☒

74.57%

GOT IT WRONG ☐

25.43%

SOLUTION

[▶ Video Solution](#)

All the elements of Group 16 form hydrides of the type H_2E ($E = O, S, Se, Te, Po$). Their acidic character increases from H_2O to H_2Te .

Boiling point depends on the size and intermolecular forces.

As the size of the central atom increases boiling point should increase, but H_2O has greater boiling point than the other hydrides due to intermolecular hydrogen bonding.

The correct order of boiling points of 16th group hydrides is $H_2O > H_2Te > H_2Se > H_2S$. The most volatile hydride of 16th group is H_2S .

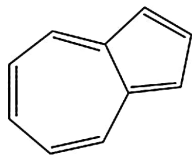
[< Prev](#)

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v

[Next >](#)

I



The compound is an example of polycyclic, non-benzenoid fused rings. Here, the aromaticity is defined by Craig's rule.

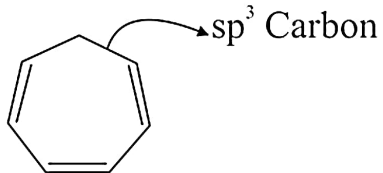
Craig's rule states that if a molecule contains $c - axis$, then the count total of number of bonds (N) can be calculated with the value of $N - 1$.

If $N - 1 = \text{even}$, the compound is aromatic.

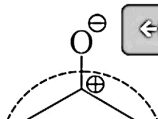
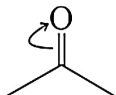
Here,

$N = 5$ and $N - 1 = 4$. So, the compound is aromatic

II



- The sp^3 carbon atom restricts the configuration of πe^- . Hence, the structure is non aromatic.



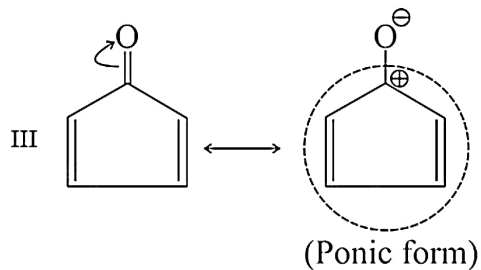
← Prev

38

✓

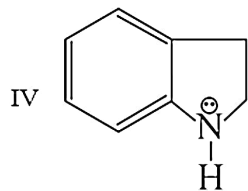
Next →

- The sp^3 carbon atom restricts the configuration of πe^- . Hence, the structure is non aromatic.



- Does not follow Huckel's rule.
- Cyclic structure (yes)
- Planer structure (yes)
- $(4n+2)\pi e^-$ (No)

The structures have $(4n\pi e^-)$ where $n = 1$. So, the structure is antiaromatic.



←Prev

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∨

Next →

The Indole structure has $10\pi e^-$, and followed Huckel's rule.

$$(4n+2) = 10$$

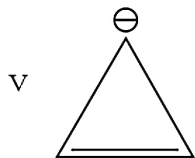
The Indole structure has $10\pi e^-$, and followed Huckel's rule.

$$(4n+2)\pi e^- = 10$$

$$4n+2 = 10$$

$$n = 2$$

So, the structure is aromatic.



Counting πe^- , we get $4\pi e^-$, and hence, the structure is antiaromatic.

- Cyclic
- Planar
- $4n\pi e^-$

DIFFICULTY

Hard

DIDN'T ATTEMPT ☐

10.59%

Others: 51 sec

CHAPTER

Hydrocarbons

GOT IT RIGHT ☒

94.10%

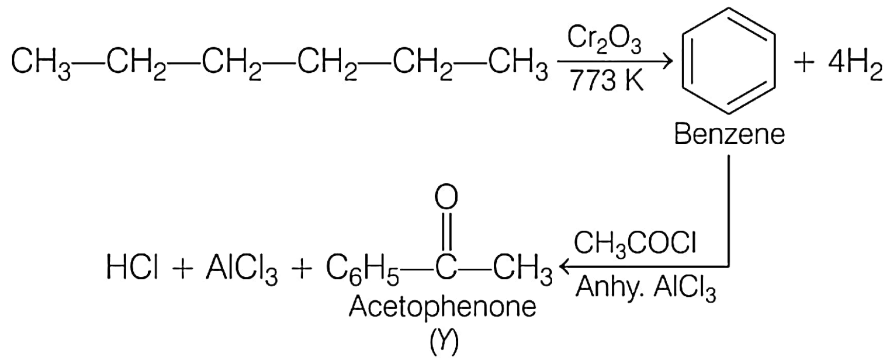
TOPIC

Reactions of aromatic compounds

GOT IT WRONG ☒

5.90%

SOLUTION

[▶ Video Solution](#)[←Prev](#)

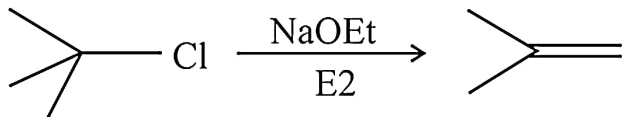
39

v

[Next→](#)

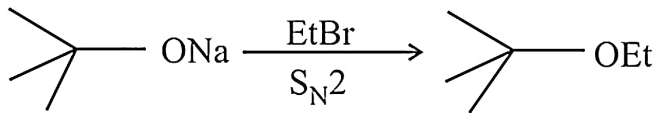
P: NaOEt

P is dehydrohalogenation of haloalkanes to give alkenes in the presence of a strong base like NaOEt.



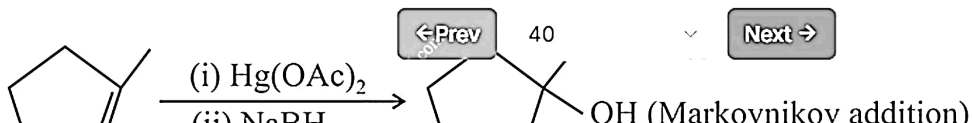
Q: EtBr

Q is Williamson ether synthesis which is an organic reaction, forming an ether from an organohalide (EtBr) and an alkoxide. Thus, Q matches with EtBr which is an organohalide.



R: Hg(OAc)₂; NaBH₄

In R, the reagent used to produce alcohol from alkene by Markovnikov's rule is Hg(OAc)₂; NaBH₄



← Prev

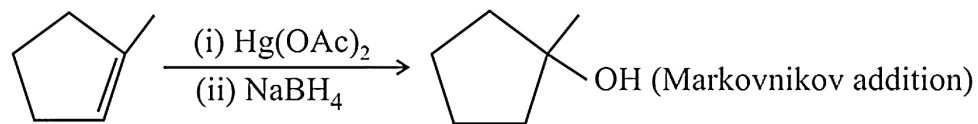
40

▼

Next →

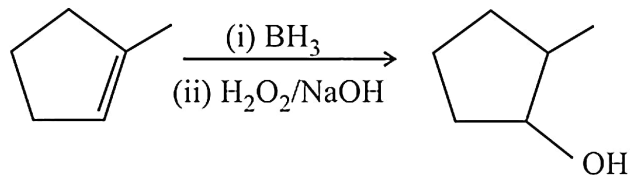
R: $\text{Hg}(\text{OAc})_2$; NaBH_4

In R, the reagent used to produce alcohol from alkene by Markovnikov's rule is $\text{Hg}(\text{OAc})_2$; NaBH_4

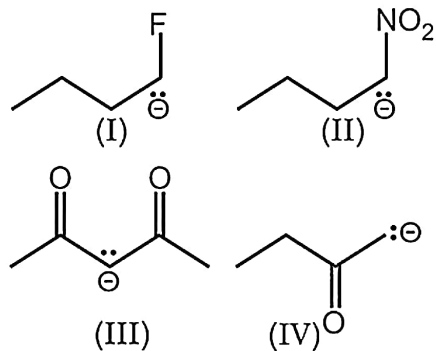


S: BH_3 ; $\text{H}_2\text{O}_2/\text{NaOH}$

S is hydroboration oxidation which is an anti-Markovnikov's reaction, which converts an alkene into alcohol. The process results in the syn-addition of hydrogen and anti-addition of a hydroxyl group, where the double bond has been. The reagent used is BH_3 ; $\text{H}_2\text{O}_2/\text{NaOH}$.



Carbanions formed after decarboxylation (loss of CO_2) are



In (I) F is electron withdrawing (-I) effect but no resonance.

In (II) $-\text{NO}_2$ is having (-I) effect and negative charge resonates with $-\text{NO}_2$

In (IV) negative charge of carbanion resonates with carbonyl but $>\text{C}=\text{O}$ is having lower (-I) effect than $-\text{NO}_2$ IV is less stable than II

In (III) the negative charge of carbanion resonates with two $>\text{C}=\text{O}$ (carbonyl groups) on both sides i.e. most stable

$\text{I} < \text{IV} < \text{II} < \text{III}$ is the order of stabilities of carbanions & same is order of reactivities towards thermal decarboxylations.

DIFFICULTY
Medium

DIDN'T ATTEMPT ☐
37.73%

Others: 34 sec

CHAPTER
Redox Reactions

GOT IT RIGHT ☒
59.69%

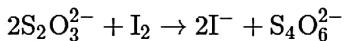
TOPIC
Redox titration

GOT IT WRONG ☐
40.31%

SOLUTION

[▶ Video Solution](#)

Titration of iodine (I_2) by thiosulphate solution ($S_2O_3^{2-}$) follows the reaction:



In the presence of starch, free iodine present in the solution imparts blue colour to the solution.

End point marks the completion of reaction, that is, at this point all the iodine present in the solution is reduced to iodide ions by thiosulphate solution.

So, due to the absence of free iodine in the solution, the solution turns colourless.

Thus, the end point is indicated by colour change in the solution from blue to colourless.

[< Prev](#)

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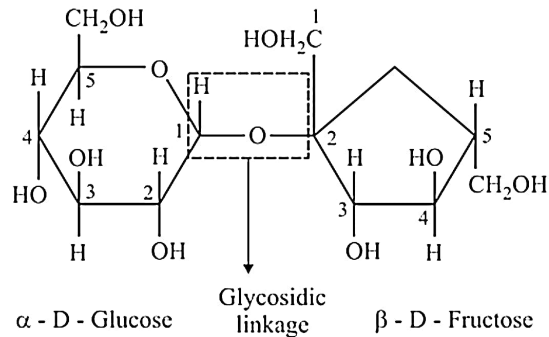
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[Next >](#)

SOLUTION

[▶ Video Solution](#)

In sucrose two monosaccharides are joined together by an oxide linkage formed by loss of water molecule. Such linkage through oxygen atom is called glycosidic linkage. In sucrose linkage is between C_1 of α -glucose and C_2 of β -fructose. Since the reducing group of glucose & fructose are involved in glycosidic bond formation, sucrose is non reducing sugar.


 $C_1\alpha - D - \text{glucose} + C_2\beta - D - \text{fructose}$

DIFFICULTY

Easy

CHAPTER

Haloalkanes and Haloarenes

TOPIC

Preparation and Properties of Monohalides

DIDN'T ATTEMPT ☐

3.60%

GOT IT RIGHT ☒

94.73%

GOT IT WRONG ☐

5.27%

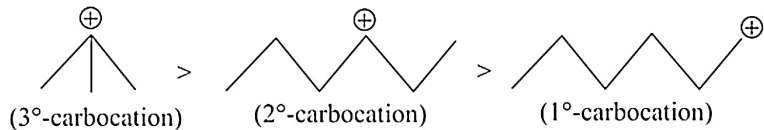
SOLUTION

[▶ Video Solution](#)

The Correct order of reactivity is (i) > (ii) > (iii) .

S_N^1 reaction occurs in two steps. In step (i) carbocation and nucleophile is produced. The carbocation thus formed is then attacked by nucleophile in step (ii).

Rate of reaction depends on the stability of carbocation, greater the stability of carbocation, greater will be its ease of formation from alkyl halide. 3° alkyl halide undergo S_N^1 reaction very fast because of the high stability of 3° carbocations, followed by 2° then 1° carbocation.

[< Prev](#)

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v

[Next >](#)

A **Correct**
49.78% marked this

B
30.97% marked this

C **Any of these**
13.87% marked this

D **none of these**
5.38% marked this

EVALUATION

[Marked for Review](#)

TIME TAKEN

You: 2 sec
Others: 26 sec

TIME ALLOTTED

90 sec

DIFFICULTY

Medium

CHAPTER

Practical Chemistry

TOPIC

Identification of Basic Radicals

DIDN'T ATTEMPT ☐

49.16%

GOT IT RIGHT ☒

49.78%

GOT IT WRONG ☒

50.22%

SOLUTION

[▶ Video Solution](#)

Qualitative distinction of ZnSO_4 and $\text{Al}_2(\text{SO}_4)_3$ is done using $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$.

[< Prev](#)

45



[Next >](#)

DIDN'T ATTEMPT ☐

15.22%

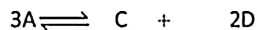
GOT IT RIGHT ☒

90.46%

GOT IT WRONG ☐

9.54%

SOLUTION

[▶ Video Solution](#)

Initially: C

at eq: C - x $\frac{x}{3}$ $\frac{2x}{3}$

$$\text{Now, } \frac{2x}{3} = 2 \times (C - x)$$

$$x = 3C - 3x$$

$$C = \frac{4}{3}x$$

$$K_c = \frac{[C][D]^2}{[A]^3}$$

$$K_c = \frac{\frac{x}{3} \times \left(\frac{2x}{3}\right)^2}{\left(\frac{4}{3}x - x\right)^3} = \frac{\left(\frac{4}{27}\right)}{\left(\frac{1}{27}\right)} = 4$$

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v

[Next >](#)

DIDN'T ATTEMPT 

33.89%

GOT IT RIGHT 

43.40%

GOT IT WRONG 

56.60%

SOLUTION

 Video Solution

$$PV^n = \text{const}$$

$$C = \frac{R}{\gamma - 1} + \frac{R}{1 - n}$$

$$C = \frac{R}{\frac{5}{3} - 1} + \frac{R}{1 - \frac{5}{2}}$$

$$C = \frac{3R}{2} - \frac{2R}{3} = \frac{5R}{6}$$

$$Q = nC\Delta T$$

$$Q = \frac{5}{6} \times \frac{R}{2} \times 36$$

$$Q = 60\text{cal}$$

 Prev

47

v

Next 

DIDN'T ATTEMPT 

51.18%

GOT IT RIGHT 

84.75%

GOT IT WRONG 

15.25%

SOLUTION

[▶ Video Solution](#)

∴ According to van't-Hoff volume (i)

$$(i) = \frac{\text{Molecular mass (calculated)}}{\text{Molecular mass (experimental)}}$$

Given, molecular mass (experimental) is double.

$$\therefore i = \frac{1}{2}$$

Also,

$$\therefore (i) = (1 - \alpha) + \frac{\alpha}{n}$$

where, α = degree of association and n = number of particles associated.

(Here, $n = 2$, ∴ $i = 1/2$)

Therefore,

$$\frac{1}{2} = (1 - \alpha) + \frac{\alpha}{2}$$

$$\frac{1}{2} = 1 - \frac{\alpha}{2}$$

$$\text{or, } \frac{\alpha}{2} = \frac{1}{2}$$

$$\text{or, } \alpha = 1$$

[< Prev](#)

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v

[Next >](#)

DIDN'T ATTEMPT 

29.21%

GOT IT RIGHT 

61.08%

GOT IT WRONG 

38.92%

SOLUTION

[▶ Video Solution](#)

$$\frac{k_1}{k_2} = \frac{A_1}{A_2} e^{\left(\frac{E_{a2} - E_{a1}}{1000}\right)}$$

$$100 = \frac{1}{10} e^{\left(\frac{(B_{a2} - B_{a1})}{1000}\right)}$$

$$1000 = e^{\left(\frac{(E_{a2} - E_{a1})}{1000}\right)}$$

$$\ln 10^3 = \left(\frac{B_{a2} - E_{a1}}{1000}\right)$$

$$\frac{3 \ln 10 \times 1000}{1000} = (E_{a2} - E_{a1})$$

[← Prev](#)

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v

[Next →](#)

EVALUATION

Marked for Review

TIME TAKEN

You: 1 sec

Others: 115 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Hard

CHAPTER

General Organic Chemistry

TOPIC

Quantitative analysis of organic compounds

DIDN'T ATTEMPT ☐

59.06%

GOT IT RIGHT ☒

71.76%

GOT IT WRONG ☐

28.24%

SOLUTION

[▶ Video Solution](#)

$$\text{Percentage of nitrogen} = \frac{1.4 \times \text{Normality (acid)} \times \text{Vol. of acid used}}{\text{mass of substance taken}}$$

$$\text{Mass of organic compound} = 0.2 \text{ g}$$

$$\text{Unused acid required} = 40 \text{ ml } \frac{N}{10} \text{ NaOH}$$

$$40 \text{ ml } \frac{N}{10} \text{ NaOH} = 40 \text{ ml } \frac{N}{10} \text{ H}_2\text{SO}_4 \equiv 20 \text{ ml } \frac{N}{5} \text{ H}_2\text{SO}_4$$

$$\text{Acid used for absorption of ammonia} = (60 - 20) \text{ ml } \frac{N}{5} \text{ H}_2\text{SO}_4 = 40 \text{ ml } \frac{N}{5} \text{ H}_2\text{SO}_4$$

$$\% \text{ of nitrogen} = \frac{1.4 \times N_1 \times V}{w} = \frac{1.4 \times \frac{1}{5} \times 40}{0.2} = 56\%.$$

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v

[Next >](#)

EVALUATION

Not Answered

TIME TAKEN

You: 17 sec

Others: 72 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Medium

CHAPTER

Complex Number

TOPIC

Conjugate, modulus and argument

DIDN'T ATTEMPT ☐

49.85%

GOT IT RIGHT ☒

32.15%

GOT IT WRONG ☒

67.85%

SOLUTION

[▶ Video Solution](#)

It is given that $|z| = 1$ which implies that $z = \cos \theta + i \sin \theta, 0 < \theta < 2\pi$:

$$\left| \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \right| = 1$$

$$\Rightarrow 2|\cos 2\theta| = 1$$

$$\Rightarrow \cos 2\theta = \frac{1}{2} \text{ or } \cos 2\theta = -\frac{1}{2}$$

$$\text{Now } \cos 2\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\text{and } \cos 2\theta = -\frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}.$$

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v

[Next >](#)

Not AnsweredYou: 59 sec
Others: 214 sec

150 sec

DIFFICULTY
EasyCHAPTER
DeterminantsTOPIC
Expansion of DeterminantsDIDN'T ATTEMPT ☐
38.67%GOT IT RIGHT ☒
91.81%GOT IT WRONG ☐
8.19%

SOLUTION

[▶ Video Solution](#)

$$\because |A| = 3$$

$$B = \begin{bmatrix} a_{11} & \frac{1}{2} a_{12} & \frac{1}{4} a_{13} \\ 2a_{21} & a_{22} & \frac{1}{2} a_{23} \\ 4a_{31} & 2a_{32} & a_{33} \end{bmatrix}$$

$$|B| = 8 \begin{vmatrix} a_{11} & \frac{1}{2} a_{12} & \frac{1}{4} a_{13} \\ a_{21} & \frac{1}{2} a_{22} & \frac{1}{4} a_{23} \\ a_{31} & \frac{1}{2} a_{32} & \frac{1}{4} a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = |A| = 3$$

$$\left| (B^T)^{-1} \right| = \frac{1}{|B^T|} = \frac{1}{|B|} = \frac{1}{3}$$

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v

[Next →](#)Activate Windows
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EVALUATION



TIME TAKEN

You: 41 sec

Others: 92 sec

TIME ALLOTTED

210 sec

DIFFICULTY

Hard

CHAPTER

Complex Number

TOPIC

Geometry of Complex Number

DIDN'T ATTEMPT ☐

53.30%

GOT IT RIGHT ☒

60.83%

GOT IT WRONG ☐

39.17%

SOLUTION

Video Solution

The image of z in the real axis is $\bar{z}$.

$\therefore$ The image is given by $\arg\left(\frac{\bar{z}-3}{\bar{z}-i}\right) = \frac{\pi}{6}$. But $\arg \bar{z} = -\arg z$

$$\therefore \arg\left(\frac{\bar{z}-3}{\bar{z}-i}\right) = \frac{\pi}{6} \Rightarrow \arg\left(\frac{z-3}{z+i}\right) = -\frac{\pi}{6} \arg\left(\frac{z+i}{z-3}\right) = \frac{\pi}{6}$$

Prev

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Next

Given $A - I = xA^2 + yA + zI$

$$A^{-1} = I = xA^3 + yA^2 + zA \quad \dots (1)$$

$$A^2 = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 6 & 4 & 5 \\ 4 & 3 & 4 \\ 5 & 4 & 6 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 6 & 4 & 5 \\ 4 & 3 & 4 \\ 5 & 4 & 6 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 21 & 15 & 20 \\ 15 & 11 & 15 \\ 20 & 15 & 21 \end{bmatrix}$$

Now using $xA^3 + yA^2 + zA = I$

$$\begin{bmatrix} 21x & 15x & 20x \\ 15x & 11x & 15x \\ 20x & 15x & 21x \end{bmatrix} + \begin{bmatrix} 6y & 4y & 5y \\ 4y & 3y & 4y \\ 5y & 4y & 6y \end{bmatrix} + \begin{bmatrix} 2z & z & z \\ z & z & z \\ z & z & 2z \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Comparing the diagonal elements

$$21x + 6y + 2z = 1 \quad \dots (1)$$

$$11x + 3y + z = 1 \quad \dots (2)$$

$$21x + 6y + 2z = 1 \quad \dots (3)$$

$$2 \times (2) - (1) \text{ gives } x = 1$$

$$3y + z = -10 \quad \dots (4)$$

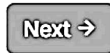
Comparing, a12

$$15x + 4y + z = 0$$

$$4y + z = -15 \quad \dots (5)$$

$$\text{for (4) \& (5) } y = -5 \text{ and } z = 5$$

$$x + y + z = 1 + (-5) + 5 = 1$$





You: 284 sec
Others: 136 sec

150 sec

Medium

Sequences and Series

Inequalities

DIDN'T ATTEMPT ☐

46.80%

GOT IT RIGHT ☒

92.92%

GOT IT WRONG ☐

7.08%

SOLUTION

[▶ Video Solution](#)

$$\begin{aligned} & \left(\frac{a^4 + a^2 + 1}{a^2} \right) \left(\frac{b^4 + 7b^2 + 1}{b^2} \right) \left(\frac{c^4 + 11c^2 + 1}{c^2} \right) \\ \Rightarrow & \left(a^2 + \frac{1}{a^2} + 1 \right) \left(b^2 + \frac{1}{b^2} + 7 \right) \left(c^2 + \frac{1}{c^2} + 11 \right) \\ & \left[\left(a - \frac{1}{a} \right)^2 + 3 \right] \left[\left(b - \frac{1}{b} \right)^2 + 9 \right] \left[\left(c - \frac{1}{c} \right)^2 + 13 \right] \end{aligned}$$

clearly minimum value occurs when $a^2 = b^2 = c^2 = 1$ and minimum value $= 3 \times 9 \times 13 = 351$

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DIDN'T ATTEMPT 

9.21%

GOT IT RIGHT 

95.94%

GOT IT WRONG 

4.06%

SOLUTION

[▶ Video Solution](#)

Let, the last element of set A is, $t_{200} = 2 + (200 - 1)2 = 400$

and the last element of set B is $t_{250} = 3 + (250 - 1)3 = 750$

Common elements are $\rightarrow 6, 12, 18 \dots$

We have to find the common elements upto 400

$$t_n = 6 + (n - 1)6 \Rightarrow 400 = 6 + 6n - 6 \Rightarrow n = \frac{400}{6} = 66.6 \dots$$

$$t_{66} = 6 + (66 - 1)6 = 396$$

Number of common elements = 66

$$\begin{aligned} n(A \cup B) &= n(A) + n(B) - n(A \cap B) \\ &= 200 + 250 - 66 = 384 \Rightarrow k = 384 \end{aligned}$$

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$f(x)$ has local minimum at $x = 2$

8.27% marked this



all of these

25.40% marked this

EVALUATION



TIME TAKEN

You: 40 sec

Others: 132 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Medium

CHAPTER

Application of Derivatives

TOPIC

Maxima Minima

DIDN'T ATTEMPT

24.53%

GOT IT RIGHT

52.98%

GOT IT WRONG

47.02%

SOLUTION

Video Solution

$$f(x) = (x-1)^2(x-2)^3e^x$$

$$\Rightarrow f'(x) = (x-1)(x-2)^2(x^2+2x-5)e^x$$

$f(x)$ has local maximum at $x = 1$, since $f'(x)$ changes from positive to negative. $f(x)$ has a point of inflexion at $x = 2$.

Prev

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Next

$$F(x) = \int_x^{x^2 + \frac{\pi}{3}} 2\cos^2 t dt$$

$$F'(x) = 2\cos^2\left(x^2 + \frac{\pi}{3}\right) \cdot 2x + 2\cos^2 x$$

$$F''(x) = 4\cos^2\left(x^2 + \frac{\pi}{3}\right) + 4x \cdot 2\cos\left(x^2 + \frac{\pi}{3}\right) \\ \left(-\sin\left(x^2 + \frac{\pi}{3}\right)\right) + 4(\cos x)(-\sin x) \dots (i)$$

$$\text{Given that area} = F'(a) + 2$$

$$\Leftrightarrow \int_0^a f(x) dx = F'(a) + 2$$

Differentiating with respect to a , we get

$$f(a) = F''(a)$$

Substituting $a = 0$, we get

$$\Rightarrow f(0) = F''(0)$$

$$= 4\cos^2 \frac{\pi}{3}$$

$$= 4\left(\frac{1}{4}\right) = 1$$

EVALUATION

Not Answered

TIME TAKEN

You: 3 sec

Others: 140 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Medium

CHAPTER

Definite Integration

TOPIC

Properties of Definite Integration

DIDN'T ATTEMPT ☐

43.94%

GOT IT RIGHT ☒

91.48%

GOT IT WRONG ☐

8.52%

SOLUTION

[▶ Video Solution](#)

$$I = 2 \int_0^1 [x \sin(\pi x)] dx$$

Now, $x \sin(\pi x) \in (0, 1)$

as $x \in (0, 1)$

$$\therefore [x \sin(\pi x)] = 0$$

$$\therefore I = 2 \int_0^1 0 dx = 0$$

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Sum

SOLUTION

[▶ Video Solution](#)

Given, lines $\mathbf{r} = (-\hat{\mathbf{i}} + 3\hat{\mathbf{k}}) + t(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}})$

and

$$\mathbf{r} = (3\hat{\mathbf{i}} + \hat{\mathbf{j}} - \hat{\mathbf{k}}) + s(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}})$$

Shortest distance between lines is

$$SD = \frac{((3+1)\hat{\mathbf{i}} + \hat{\mathbf{j}} - (\hat{\mathbf{k}} + 3)\hat{\mathbf{k}}) \cdot [(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}) \times (2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}})]}{|(2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + 6\hat{\mathbf{k}}) \times (2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}})|}$$

$$SD = \frac{(4\hat{\mathbf{i}} + \hat{\mathbf{j}} - 4\hat{\mathbf{k}}) \cdot (12\hat{\mathbf{j}} + 6\hat{\mathbf{i}})}{|12\hat{\mathbf{i}} + 8\hat{\mathbf{j}} - 8\hat{\mathbf{k}}|} \times |(2\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{j}} - 8\hat{\mathbf{k}})|$$

$$SD = \frac{48 + 8 + 32}{\sqrt{144 + 64 + 64}} = \frac{88}{4\sqrt{17}} = \frac{22}{\sqrt{17}}$$

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DIFFICULTY

Medium

CHAPTER

Differential Equations

TOPIC

Linear Differential Equation

DIDN'T ATTEMPT 

40.05%

GOT IT RIGHT 

94.26%

GOT IT WRONG 

5.74%

SOLUTION

 Video Solution

$$2xy \frac{dy}{dx} = y^2 + x \ln x$$

$$2y \frac{dy}{dx} - \frac{y^2}{x} = \ln x$$

$$\text{Put, } y^2 = t \Rightarrow 2y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} - \frac{t}{x} = \ln x$$

$$\text{I.F.} = e^{\int \frac{-1}{x} dx} = e^{-\ln x} = \frac{1}{x}$$

$$\text{The solution is } \frac{t}{x} = \int \frac{\ln x}{x} dx$$

$$\frac{t}{x} = \frac{(\ln x)^2}{2} + c$$

$$\frac{y^2}{x} = \frac{(\ln x)^2}{2} + c$$

$$2y^2 = x(\ln x)^2 + 2cx$$

 Prev

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Next 

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$$x + y - 1 = 0$$

4.19% marked this



$$\text{line } x + y = 0$$

2.15% marked this

EVALUATION



TIME TAKEN

You: 33 sec

Others: 105 sec

TIME ALLOTTED

180 sec

DIFFICULTY

Medium

CHAPTER

Parabola

TOPIC

standard equation of parabola

DIDN'T ATTEMPT ☐

45.22%

GOT IT RIGHT ☒

90.33%

GOT IT WRONG ☐

9.67%

SOLUTION

Video Solution

Line Joining $(1, 2)$ and $(2, 1)$ and line Joining $(3, 4)$ and $(4, 3)$ are parallel.

Also, the perpendicular bisector of any two parallel chords is axis of parabola.
Hence, axis of the parabola is $x - y = 0$

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Next

$$S_1 > 0 \Rightarrow 2\cos^2\theta + 3\cos\theta - 2 > 0$$

$$(2\cos\theta - 1)(\cos\theta + 2) > 0$$

$$\cos\theta > \frac{1}{2}$$

$$\therefore \theta \in \left[0, \frac{\pi}{3}\right) \cup \left(\frac{5\pi}{3}, 2\pi\right]$$

$$\text{given } \theta = \frac{n\pi}{6} \Rightarrow \theta = \frac{\pi}{6}, \frac{11\pi}{6}, \frac{12\pi}{6}$$

$$\therefore n \in \{1, 11, 12\}$$

The given line passes through $A(5, -2, 6)$

$$(PQ)^2 = (AP)^2 - (AQ)^2$$

$$(AP)^2 = (4 - 5)^2 + (-5 + 2)^2 + (3 - 6)^2 = 19$$

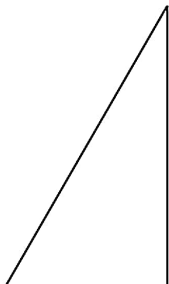
AQ is the projection of AP on the given line

$$\therefore AQ = (5 - 4) \frac{3}{\sqrt{50}} + (-2 + 5) \frac{(-4)}{\sqrt{50}} + (6 - 3) \left(\frac{-5}{\sqrt{50}} \right) = \frac{6}{\sqrt{50}}$$

$$\Rightarrow (PQ)^2 = 19 - \frac{36}{50} = \frac{457}{25}$$

$$\Rightarrow 25(PQ)^2 - 450 = 7$$

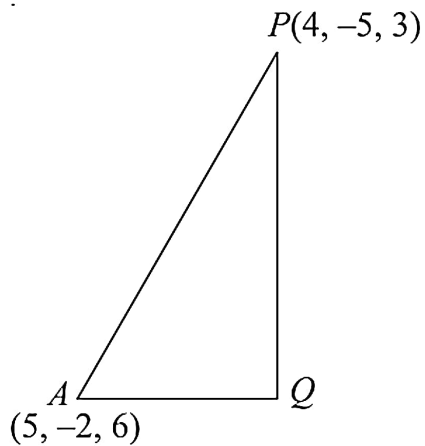
$P(4, -5, 3)$



$$\therefore AQ = (5-4) \frac{1}{\sqrt{50}} + (-2+5) \frac{1}{\sqrt{50}} + (6-3) \left(\frac{1}{\sqrt{50}} \right) = \frac{4}{\sqrt{50}}$$

$$\Rightarrow (PQ)^2 = 19 - \frac{36}{50} = \frac{457}{25}$$

$$\Rightarrow 25(PQ)^2 - 450 = 7$$



$P.V$ of $M((\vec{r} = \vec{a} + \lambda \vec{b}))$

$$\overrightarrow{BM} = (\vec{a} + (\lambda - 1)\vec{b})$$

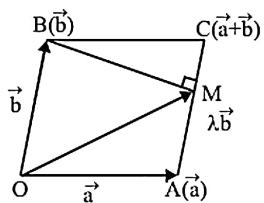
$$\overrightarrow{BM} \cdot \overrightarrow{AC} = 0$$

$$(\vec{a} + (\lambda - 1)\vec{b}) \cdot \vec{b}$$

$$\vec{a} \cdot \vec{b} + (\lambda - 1)(\vec{b})^2 = 0$$

$$4(\lambda - 1) = -1$$

$$\lambda - 1 = -\frac{1}{4}$$



$$\lambda = \frac{3}{4}$$

$$\overrightarrow{BM} = \left| \vec{a} + \frac{3}{4}\vec{b} - \vec{b} \right|$$

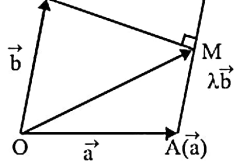
$$\overrightarrow{BM} = \left| \vec{a} - \frac{\vec{b}}{4} \right|^2 = |\vec{a}|^2 + \frac{|\vec{b}|^2}{16} - \frac{2 \cdot \vec{a} \cdot \vec{b}}{4}$$

← Prev

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✓

Next →



$$\lambda = \frac{3}{4}$$

$$\overrightarrow{BM} = \left| \vec{a} + \frac{3}{4}\vec{b} - \vec{b} \right|$$

$$\overrightarrow{BM} = \left| \vec{a} - \frac{\vec{b}}{4} \right|^2 = |\vec{a}|^2 + \frac{|\vec{b}|^2}{16} - \frac{2 \cdot \vec{a} \cdot \vec{b}}{4}$$

$$= 4 + \frac{1}{4} - \frac{1}{2} = \frac{16 + 1 - 2}{4} = \frac{15}{4}$$

$$|\overrightarrow{BM}|^2 = \frac{15}{4}$$

$$|\overrightarrow{BM}| = \frac{\sqrt{15}}{2}$$

SOLUTION

[▶ Video Solution](#)

$$\bar{x} = \frac{\sum x_i}{5} \Rightarrow \sum x_i = 5 \times 2 = 10$$

$$\sigma_x^2 = \frac{\sum x_i^2}{5} - \bar{x}^2 \Rightarrow 4 + 2^2 = \frac{\sum x_i^2}{5} \Rightarrow \sum x_i^2 = 40$$

$$\bar{y} = \frac{\sum y_i}{5} \Rightarrow \sum y_i = 5 \times 4 = 20$$

$$\sigma_y^2 = \frac{\sum y_i^2}{5} - \bar{y}^2 \Rightarrow \frac{\sum y_i^2}{5} = 5 + 4^2 = 21 \Rightarrow \sum y_i^2 = 105$$

$$\bar{x} = \frac{\sum (x_i + y_i)}{10} = \frac{30}{10} = 3$$

$$\sigma^2 = \frac{\sum x_i^2 + y_i^2}{10} - 3^2 = \frac{145}{10} - 9 = 14.5 - 9 = 5.5$$

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Let A reach place of meeting at 7 hours x minutes and B reaches place of meeting at 7 hours y minutes.

Then,

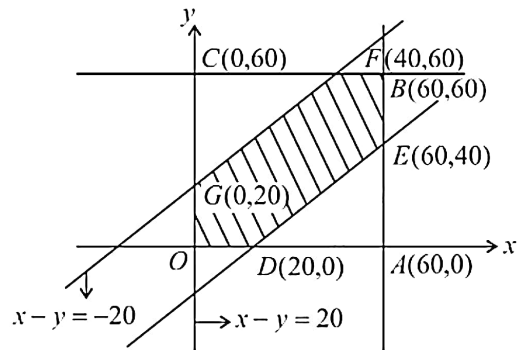
$$0 \leq x \leq 60$$

$$0 \leq y \leq 60$$

Condition for meeting to happen.

$$|x - y| \leq 20$$

Then,



Total area

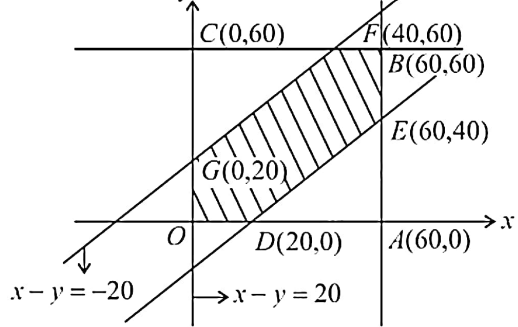
$$= \text{ar}(OABCO) = 60 \times 60 = 3600$$

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∨

Next →



Total area

$$= \text{ar}(OABCO) = 60 \times 60 = 3600$$

Favourable area

$$= \text{ar}(ODBEFG)$$

$$= \text{ar}(OABCO) - (A_1 + A_2)$$

$$= 3600 - \left\{ \frac{1}{2} \times 40 \times 40 + \frac{1}{2} \times 40 \times 40 \right\}$$

$$= 2000$$

Hence, required probability is

$$= \frac{2000}{3600} = \frac{5}{9}$$

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∨

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EVALUATION



TIME TAKEN

You: 151 sec
Others: 164 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Easy

CHAPTER

Inverse Trigonometric Functions

TOPIC

Infinite Series

DIDN'T ATTEMPT

47.98%

GOT IT RIGHT

79.21%

GOT IT WRONG

20.79%

SOLUTION

[▶ Video Solution](#)

$$\begin{aligned}y &= \tan^{-1} \frac{1}{1+x+x^2} + \tan^{-1} \frac{1}{x^2+3x+3} + \dots + 2n \text{ terms} \\&= \tan^{-1} \frac{(x+1)-x}{1+x(1+x)} + \tan^{-1} \frac{(x+2)-(x+1)}{1+(x+1)(x+2)} + \dots (2n \text{ terms}) \\&= \tan^{-1}(x+1) - \tan^{-1}x + \tan^{-1}(x+2) - \tan^{-1}(x+1) + \dots + \tan^{-1}(x+2n) - \tan^{-1}(x+(2n-1)) \\&= \tan^{-1}(x+2n) - \tan^{-1}x \\y(0) &= \tan^{-1}(2n)\end{aligned}$$

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SOLUTION

[▶ Video Solution](#)

Since $\sin^2 18^\circ$ and $\cos^2 36^\circ$ are the roots of a quadratic equation.

∴ Sum of roots = $\sin^2 18^\circ + \cos^2 36^\circ$

$$= \left(\frac{\sqrt{5}-1}{4}\right)^2 + \left(\frac{\sqrt{5}+1}{4}\right)^2$$

$$= \frac{5+1-2\sqrt{5}}{16} + \frac{5+1+2\sqrt{5}}{16}$$

$$= \frac{12}{16} = \frac{3}{4}$$

and product of roots = $\sin^2 18^\circ \cdot \cos^2 36^\circ$

$$= \left(\frac{\sqrt{5}-1}{4}\right)^2 \left(\frac{\sqrt{5}+1}{4}\right)^2$$

$$= \left(\frac{5-1}{4 \times 4}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

Required equation whose roots are $\sin^2 18^\circ$ and $\cos^2 36^\circ$, is

$$x^2 - (\text{sum of roots})x + (\text{product of roots}) = 0$$

$$\Rightarrow x^2 - \frac{3}{4}x + \frac{1}{16} = 0$$

$$\Rightarrow 16x^2 - 12x + 1 = 0$$

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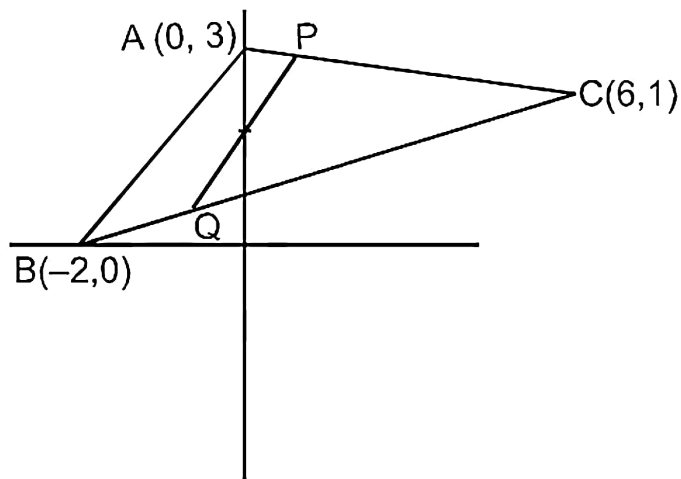
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Since $(\lambda, \lambda + 1)$ lies on $y = x + 1$

equation of $AB : 3x - 2y + 6 = 0$; $BC : x - 8y + 2 = 0$; $AC : x + 3y - 9 = 0$



Line $y = x + 1$ cuts AC at $P\left(\frac{3}{2}, \frac{5}{2}\right)$ cut BC at $Q\left(\frac{-6}{7}, \frac{1}{7}\right)$. Hence $\lambda \in \left(\frac{-6}{7}, \frac{3}{2}\right)$

DIDN'T ATTEMPT ☐

46.06%

GOT IT RIGHT ☒

46.26%

GOT IT WRONG ☐

53.74%

SOLUTION

[▶ Video Solution](#)

$$\frac{(e^x - 1)(2 - \sin x)(x^2 - 5x + 4)}{x^2(x^2 - x + 2)(2x + 3)} \leq 0$$

We can observe that $2 - \sin x > 0$ and $x^2 - x + 2 > 0 \forall x \in \mathbb{R}$

so, the wavy curve is

$$x \in \left(-\frac{3}{2}, 0\right) \cup [1, 4]$$

So, the integral values of x are $-1, 1, 2, 3, 4$

Hence, $\lambda = 5$.

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DIDN'T ATTEMPT 

45.07%

GOT IT RIGHT 

85.22%

GOT IT WRONG 

14.78%

SOLUTION

[▶ Video Solution](#)

$$\therefore A^T |A| B = A |B| B^T$$

Taking determinant on both sides, we get,

$$|A^T| |A| |B| = |A| |B| |B^T|$$

$$\Rightarrow |A| = 2$$

$$\text{Now, } \left| AB^{-1} \text{adj}(A^T B)^{-1} \right|$$

$$|A| \times \frac{1}{|B|} \times \frac{1}{|\text{adj}(A^T B)|}$$

$$|A| \times \frac{1}{|B|} \times \frac{1}{|A^T B|^2} = \frac{|A|}{|B|^3 |A|^2} = \frac{1}{16}$$

$$\text{i.e. } 4K = \frac{1}{4} = 0.25$$

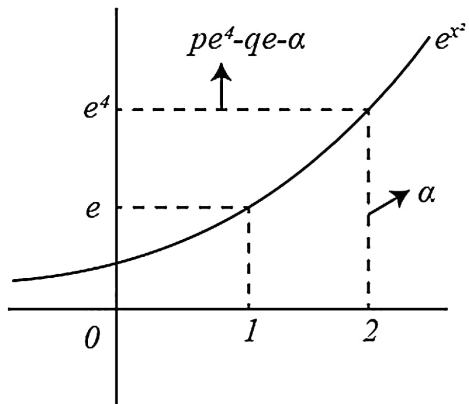
$$16K = 1$$

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$$y = e^{x^2} \rightarrow \ln y = x^2 \Rightarrow x = \sqrt{\ln y}$$

$$\int_1^2 e^{x^2} \cdot dx + \int_e^{e^4} \sqrt{\ln x} \cdot dx = \phi + pe^4 - qe - \phi$$

$$2e^4 - e = pe^4 - qe$$

$$p = 2; q = 1 \Rightarrow p + q = 3$$

Given, $f(1) = \frac{1}{3}$ and $6 \int_1^x f(t) dt = 3xf(x) - x^3$ for all $x \geq 1$

Using Newton-Leibniz formula.

Differentiating both sides,

$$\Rightarrow 6f(x) \cdot 1 - 0 = 3f(x) + 3xf'(x) - 3x^2$$

$$\Rightarrow 3xf'(x) - 3f(x) = 3x^2$$

$$\Rightarrow f'(x) - \frac{1}{x}f(x) = x$$

$$\Rightarrow \frac{xf'(x) - f(x)}{x^2} = 1$$

$$\Rightarrow \frac{d}{dx} \left\{ \frac{f(x)}{x} \right\} = 1$$

Integrating both sides,

$$\Rightarrow \frac{f(x)}{x} = x + C \quad \left[\because f(1) = \frac{1}{3} \right]$$

$$\frac{1}{3} = 1 + C$$

$$\Rightarrow C = -\frac{2}{3}$$

$$f(x) = x^2 - \frac{2}{3}x$$

$$\Rightarrow f(2) = 4 - \frac{4}{3} = \frac{8}{3}$$

$$\therefore 3f(2) = 8$$

EVALUATION



TIME TAKEN

You: 22 sec

Others: 151 sec

TIME ALLOTTED

150 sec

DIFFICULTY

Medium

CHAPTER

Binomial Theorem

TOPIC

Remainder and Divisibility Problems

DIDN'T ATTEMPT ☐

14.88%

GOT IT RIGHT ☒

86.39%

GOT IT WRONG ☐

13.61%

SOLUTION

[▶ Video Solution](#)

We have, $7^{103} = 7(49)^{51} = 7(50 - 1)^{51}$
 $= 7(50^{51} - {}^{51}C_1 50^{50} + {}^{51}C_2 50^{49} - \dots - 1)$
 $= 7(50^{51} - {}^{51}C_1 50^{50} + {}^{51}C_2 50^{49} - \dots) - 7 + 18 - 18$
 $= 7(50^{51} - {}^{51}C_1 50^{50} + {}^{51}C_2 50^{49} - \dots) - 25 + 18$
 $= k + 18$ (say) where k is divisible by 25,
 $\therefore$ remainder is 18.

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